Appliance Energy Consumption Forecasting

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1. Introduction

Short-term forecasting of electricity consumption can help in a better understanding and management of energy demand. Energy consumption by appliances is influenced by household activities and thus has temporal structure that can be utilized for energy forecasting purposes. In this project, various techniques are explored in order to forecast hourly energy consumption of the appliances.

Several benchmark methods, Mean, Naïve, Daily seasonal Naïve, weekly seasonal Naïve and Drift are compared with each other and with three primary modeling techniques – SARIMA, SARIMAX and XGBoost. In addition, the performance of the pre-trained foundation model called Chronos-T5-tiny was compared with these specialized models in order to determine its performance on the data set.

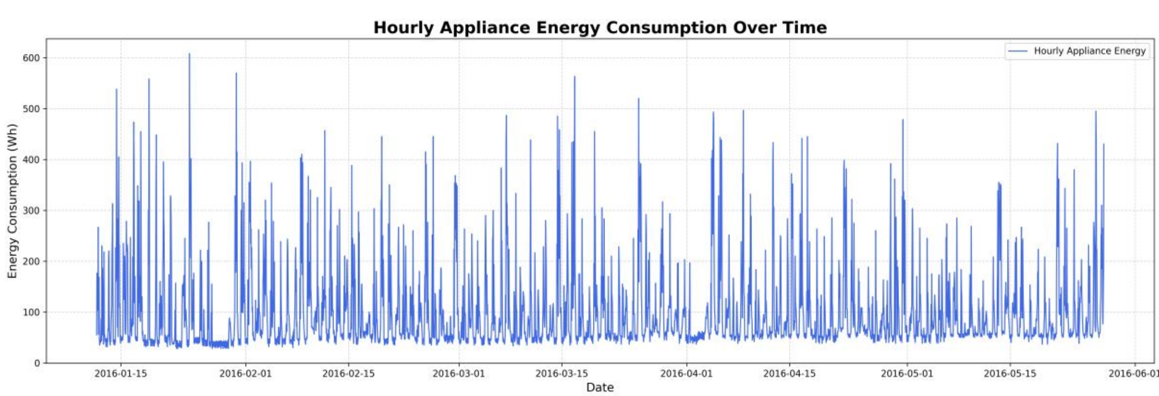
The target variable chosen was Appliances. The final dataset used in modeling had 3,290 observations and 28 features. A time series split was applied to ensure that there is no use of future data during training. In the experiment with the SARIMA model, 2,954 observations were used for training, while the last 336 observations, covering 14 days, were used for testing.

The models were assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE) and R^2. These findings indicate that SARIMA produced the best performance among all models, which implies that the inclusion of temporal dependence and daily seasonality into the model led to better results compared to the model complexity increase.

2. Data Preparation and Exploratory Analysis

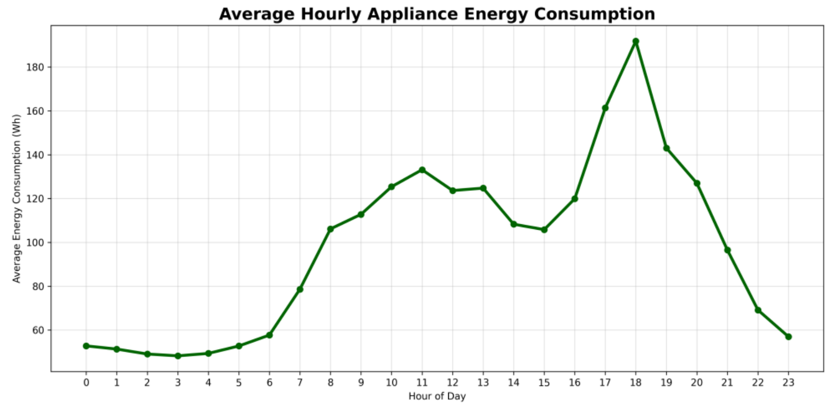
The initial observations were structured into an hourly time series, where the Appliances variable served as the forecasting target. Data preparation consisted of sorting observations in chronological order, as well as dealing with missing observations and generating further time-based and lagged variables to be used by the machine learning model. Exploratory analysis was vital since the approach chosen had to mirror the structure of the data, as opposed to just the complexity of the model.

Figure 1 – Appliance energy consumption



The data contained significant variations over time, with repetitive changes in consumption. As such, the hourly nature of the series indicated that the household behavior was not arbitrary, and that past observations could give insight into future demand.

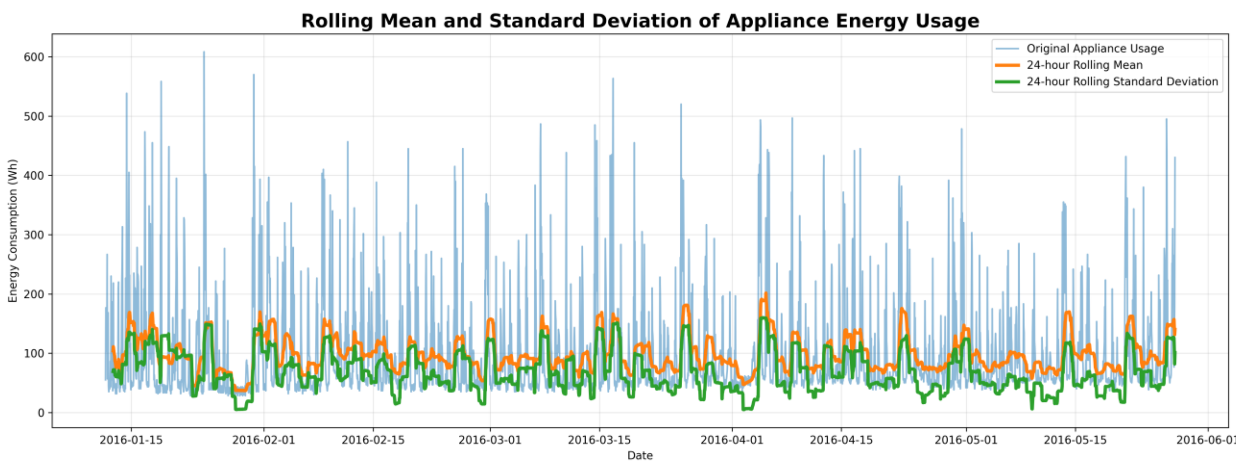
Figure 2 – Daily/hourly pattern



The repetitive daily behavior motivated modeling of the 24 observations seasonality period, which corresponds to one day since the data is hourly.

After that, stationarity was examined using rolling statistics, ADF and KPSS test results, as well as ACF and PACF plots.

Figure 3 – Stationarity diagnostics



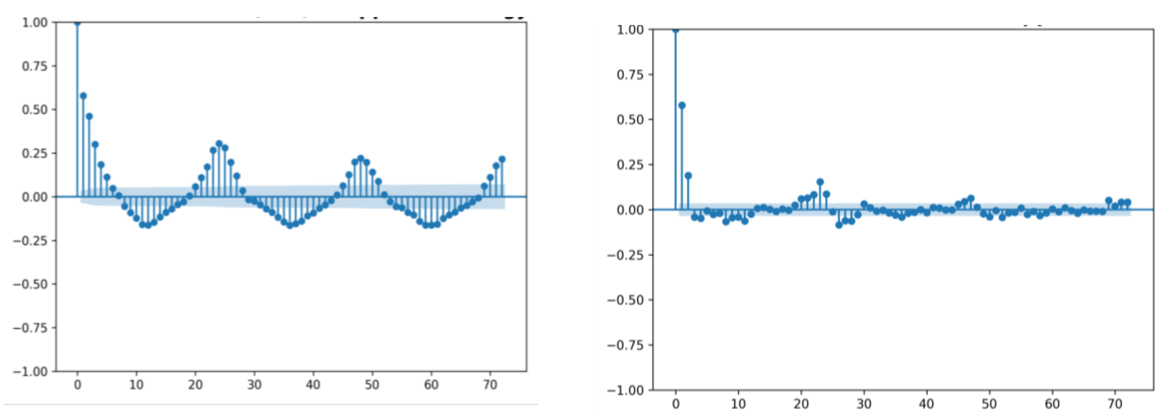


Figure: ACF and PACF autocorrelation diagnostics

This information indicates the presence of significant autocorrelation and seasonality in the time series and proves the validity of using models with explicit modeling of these features.

3. Modeling Methodology

3.1 Benchmark Models

First, simple forecasting techniques were used. Among such techniques were Mean, Naive, Daily Seasonal Naive, Weekly Seasonal Naive and Drift. The reason for using these models was to set the benchmark for other models. Only if a more complex model allows achieving better results than simple models is its development worth considering.

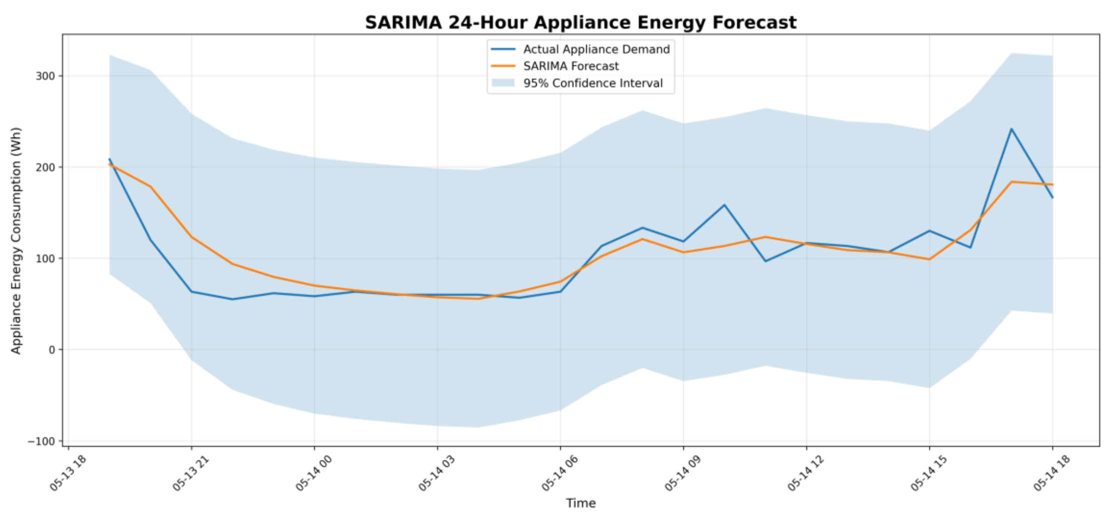
Among the models evaluated, the Weekly Seasonal Naive performed best, having achieved an MAE of 43.46 and RMSE of 81.41. The results suggest that the appliance demand has a clear recurring weekly cycle, but the high error rate as compared to SARIMA implies that simply reusing past observations is not enough to describe all temporal dependencies.

3.2 SARIMA

SARIMA was chosen due to the results of exploratory analysis, which proved the presence of temporal and daily seasonal behavior. SARIMA includes auto-regression, differencing and moving average elements, along with their seasonal counterparts.

A grid search of possible non-seasonal orders was conducted. The best combination of non-seasonal orders turned out to be: (2, 0, 6) with an AIC value of 32151.39. The chosen seasonality of 24 matches the sampling rate of one hour and daily pattern found via EDA. Hence, the developed model incorporated short-term auto-regression with the daily seasonality, rather than using only reoccurrences of historical data. The following seasonal order was the final one: (1, 1, 1, 24), where 24 refers to the number of hours in one day. In this regard, the selection of the parameters was influenced by both the statistical analysis and the daily nature of the data observed.

Figure 4 – SARIMA forecast



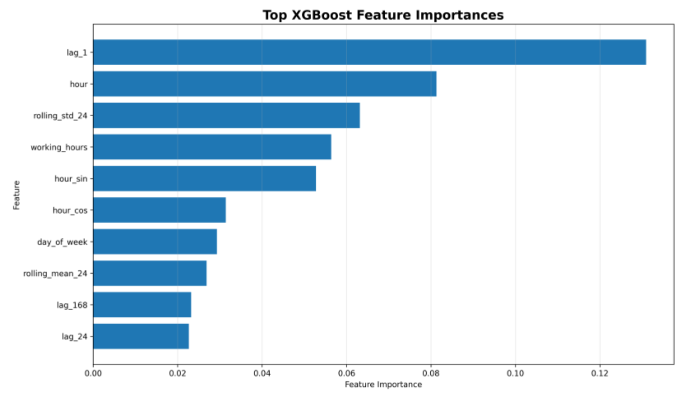
The model forecasts in general direction of the series, but specific peaks and troughs are not always captured.

3.3 SARIMAX

The SARIMAX model was employed to find out whether the inclusion of further explanatory variables would be able to enhance forecasts over the autoregressive and seasonal components modeled by the SARIMA process. However, the performance of the model was severely compromised as indicated by its poor error measures, namely MAE of 298.52, RMSE of 323.18, MAPE of 361.33%, and R^2 of −41.932. In this way, the model did not prove superior even to the best performing benchmark with seasonality, namely Weekly Seasonal Naïve (MAE 43.46; RMSE 81.41).

3.4 XGBoost

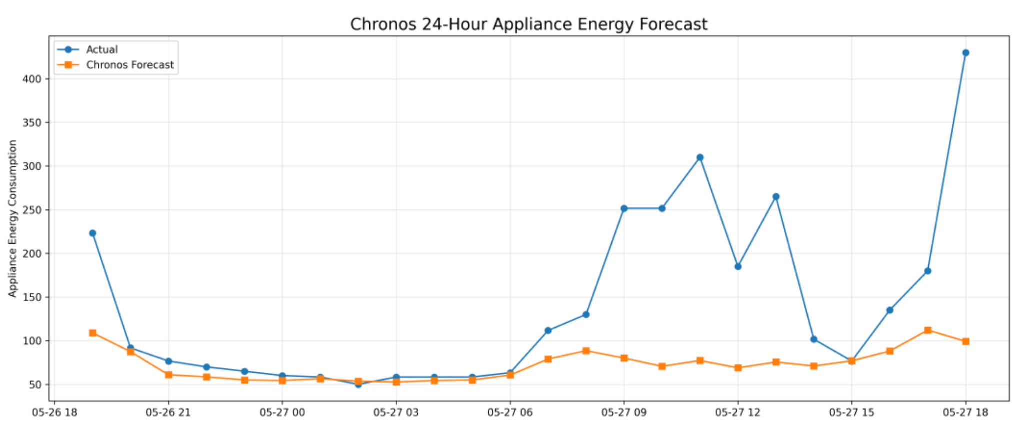
XGBoost performed better, with an MAE of 42.74 and RMSE of 51.44, compared to the benchmark algorithms, although it was significantly less successful than SARIMA. As shown in **Figure 5** The lag_1, hour, rolling_std_24, working_hours and hour_sin features had the greatest impact on the results. This suggests that the latest demand of appliances and time of day factors were more significant than many other factors. Nonetheless, the negative R^2 score of -0.088 shows that the XGBoost model failed to predict the test period variations as well as SARIMA.



3.5 Chronos-T5-tiny

The Chronos-T5-tiny model was added as a basis for comparison using foundation models. Chronos, unlike SARIMA, is developed to be a general pretrained time-series model and was not explicitly designed based on the data structure.

Figure 6 – Chronos forecast



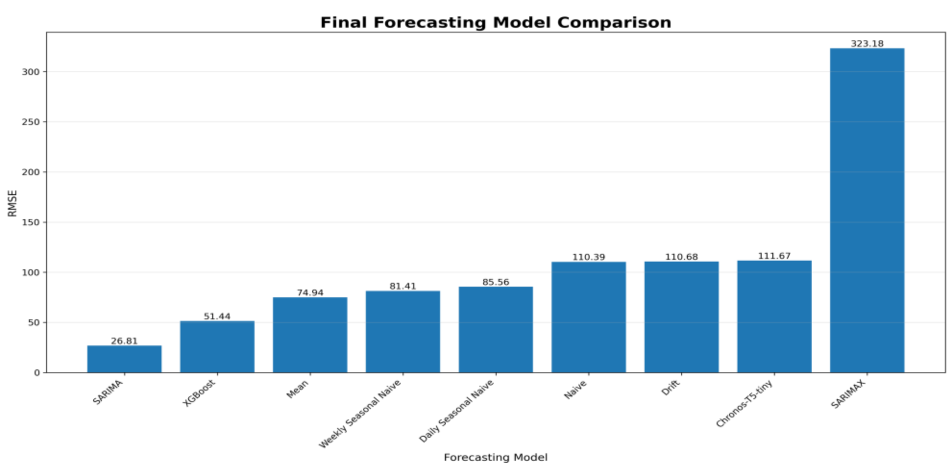
The Chronos model had a score of 67.67 in terms of MAE, 111.67 in terms of RMSE, 31.05% in MAPE, and R^2 of −0.276. Its lower performance doesn't mean that foundation models can't be used in forecasting. Rather, it shows that this particular model couldn't take full advantage of the high-level data structure. In other words, such a zero-shot foundation model is not always better than a specialized model, built specifically to fit the strong seasonality of the dataset.

4. Results and Model Comparison

Table 1 – Model evaluation

Model	MAE	RMSE	MAPE (%)	R^2
SARIMA	**18.62**	**25.78**	**19.06**	**0.727**
XGBoost	42.74	51.44	45.15	−0.088
Mean	50.26	74.94	53.70	−0.003
Weekly Seasonal Naive	43.46	81.41	37.35	−0.183
Daily Seasonal Naive	48.31	85.56	43.33	−0.307
Naive	85.55	110.39	112.91	−1.176
Drift	85.80	110.68	113.31	−1.187
Chronos-T5-tiny	67.67	111.67	31.05	−0.276
SARIMAX	298.52	323.18	361.33	−41.932

Figure 7 – Model comparison



The best performance according to the main criteria belonged to the SARIMA model. Its MAE (18.62) was 57.1% lower than that of the best benchmark Weekly Seasonal Naive (43.46) and 56.4% lower than that of XGBoost (42.74). Its RMSE (25.78) was 68.3% lower than that of Weekly Seasonal Naive (81.41) and 49.9% lower than that of XGBoost (51.44). In addition, it had the highest R^2 (0.727), which means that it could explain much more of the variation within the testing period.

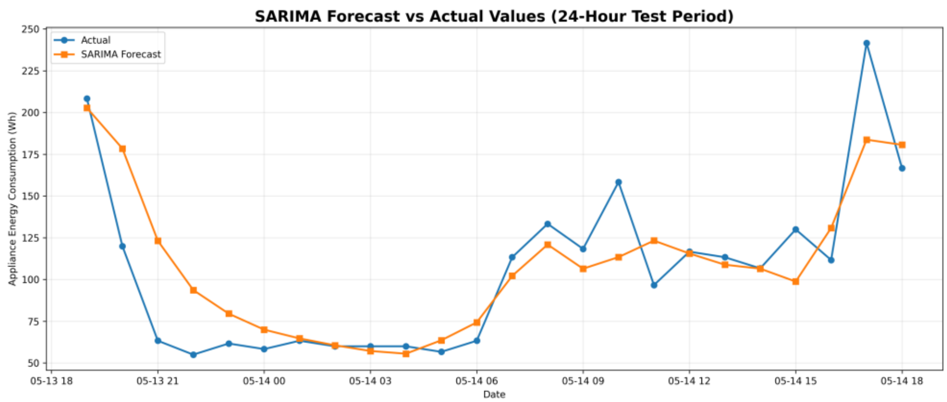
This result is especially important because SARIMA was computationally less complicated than some of the other models considered. The reason for the success of SARIMA is evidently the applicability of the model to the data structure. MAPE must, however, be taken with caution as appliance use may sometimes become quite small, making percentage errors disproportionately high. This issue applies to

SARIMAX MAPE which amounted to 361.33%. For this reason, MAE and RMSE are used as more reliable criteria to measure absolute performance in this experiment, MAPE being used as an additional criterion.

5. Forecast Analysis

The SARIMA model was tested for its performance in a forecasting horizon of 24 hours and was assessed in terms of its forecasted values against observations from the testing period.

Figure 8 – Forecast versus actual values



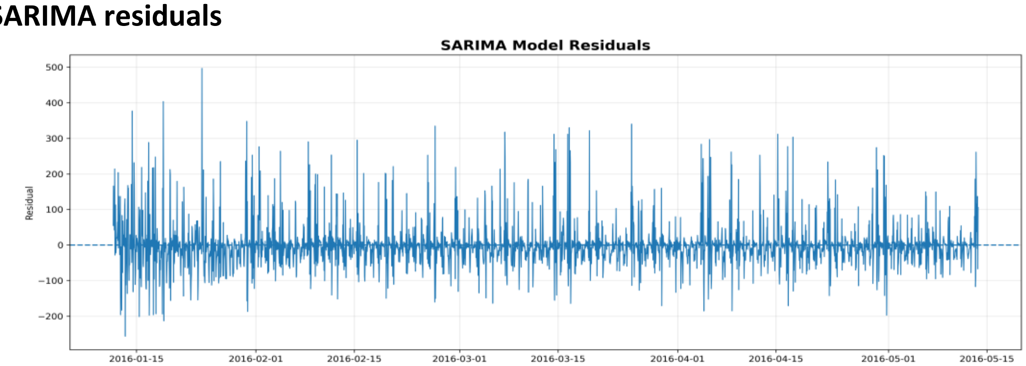
In general, the predictions matched the trend of the observed data; however, there were some instances where specific peaks and troughs were not captured. This is inevitable since appliance usage is affected by household activities which are not predictable from past data.

In the last common performance test, SARIMA model gave results of 18.62 MAE and 25.78 RMSE. One can also notice from the graph that this model was able to capture the trend of the observations, though sometimes it underestimated sharp changes in it. It means that SARIMA model is capable of capturing predictable time series structures, but not random household behavior.

6. Residual Analysis

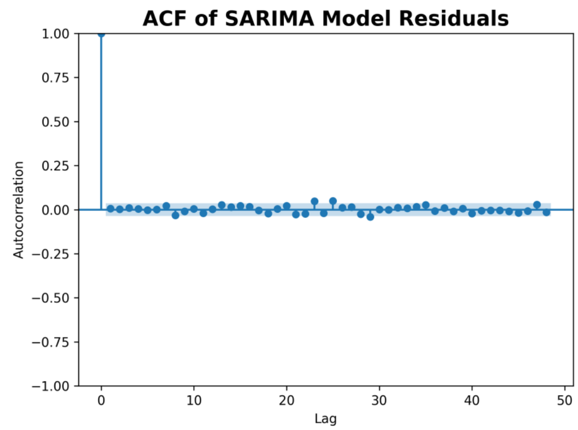
Residual analysis was conducted for the selected SARIMA model to see if there is any systematic information that is left behind.

Figure 9 – SARIMA residuals



The use of residuals in analysis is important as a good forecasting model should have residuals that are close to zero and do not exhibit any systematic pattern

Figure 10 – Residual autocorrelation



The residual ACF was examined to determine whether systematic autocorrelation remained after fitting SARIMA. Ideally, residuals should behave approximately like white noise, with no substantial remaining autocorrelation. The residual distribution was also inspected to identify skewness or unusually large errors. These diagnostics complement MAE and RMSE because a model can achieve good average accuracy while still leaving systematic structure unexplained.

7. Critical Discussion

As we can see from the results, the key feature of this data set is a combination of a robust weekly/daily seasonality and short-term autocorrelation. It is why the Weekly Seasonal Naive model showed the best results among benchmarks, and SARIMA gave even more improvement by modeling the autoregressive and seasonal components. The difference in MAE between this model and the strongest benchmark is 57.1%, and in terms of RMSE – 68.3%. We can say that it is a serious improvement rather than a slight one.

However, in case of SARIMAX, we have an example of how additional information does not necessarily improve forecasting accuracy. Both MAE and RMSE of SARIMAX (298.52 and 323.18 correspondingly) were significantly worse than those of the strongest benchmark as well as SARIMA. It means that the chosen exogenous variables or their specification cannot be used as predictors. Moreover, there is another practical drawback, which is the following: weather, humidity and indoor temperature will not be known in advance without forecasting them separately.

XGBoost achieved a decent MAE of 42.74, while its negative R^2 value means that it captured less variance than SARIMA. However, XGBoost feature importance is helpful – lag_1, hour, rolling_std_24, working_hours and hour_sin were some of the most important features. Thus, the recent consumption and the pattern of daily activities can provide more information for prediction than the mere increase in the number of features.

Chronos-T5-tiny could not surpass the specialised models as well. With MAE of 67.67 and RMSE of 111.67, it means that the zero-shot foundation model was less able to utilise the strong seasonal pattern of the dataset than SARIMA. It does not mean that foundation models are worse in general; however, in this particular case, their extra flexibility did not provide any accuracy benefits.

On the whole, the findings lend themselves to the rule that suitability matters more than complexity of the models. The SARIMA model can be described as an easily interpretable, computationally feasible model, which was very well suited for the temporal pattern that was being observed. On the other hand, XGBoost and Chronos provide more flexibility if more predictors are used.

8. Limitations and Future Improvements

Since the evaluation employed a chronological 14 days hold out period, performance may vary according to the chosen period for testing. Rolling origin validation would give more robust results for different periods.

Moreover, SARIMAX results indicate the significance of careful evaluation of the exogenous variables used. Future work can assess the presence of multicollinearity, stability of the features as well as whether the features can actually be accessed during the forecasting period. The ablation study for XGBoost can be carried out by analyzing the effect of each feature group, such as lag features, time-related features and weather/sensor features.

Future research may employ probabilistic forecasting by analyzing the prediction intervals along with metrics like MASE rather than focusing only on point forecasts and MAPE. Recently created foundation models can also be analyzed; however, their computational cost should be compared with their performance gain.

9. Conclusion

Nine models for forecasting hourly appliance energy usage were tested. The SARIMA model proved to be the most successful, with metrics of MAE = 18.62, RMSE = 25.78, MAPE = 19.06% and R^2 = 0.727. SARIMA outperformed the best benchmarking model, which is the Weekly Seasonal Naïve model, by a substantial margin, while XGBoost offered a reasonably accurate yet less precise alternative.

From the results obtained one can see that the structure of data was much more significant than the complexity of model used. SARIMA was able to use seasonal and autocorrelation structure identified during the EDA phase, while SARIMAX, XGBoost and Chronos did not offer any benefit from being more complex in the current experiment. Thus, in real-life application of smart home forecasting SARIMA would be the most balanced choice between precision, interpretability and computational efficiency.

References

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